

ANALYSTLAB AFRICA

Experience Lab Internship Programme



HealthConnect Clinic Review of Week 4 Foundation

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Project: HealthConnect Clinic Experience Lab – Improving Patient Appointment Attendance and
Healthcare Support Using Data and AI
Track: Data Science

Prepared for the AnalystLab Africa Experience Lab
Week 5 Submission – Transition Note

Review of Week 4 Foundation

Before beginning the Week 5 practical work, the Week 4 submission was reviewed. This is a brief transition note summarising that foundation, not a repetition of the full Week 4 documentation.

1. The Problem Defined

A supervised binary classification problem: predicting, using only information available at the time of booking, whether a scheduled HealthConnect appointment will result in a no-show. The model is intended to help clinic staff prioritise outreach for high-risk appointments, not to automate scheduling decisions.

2. Week 4 Proposed Approach

- Binary target (No-Show vs. Attended), with Cancelled appointments excluded from the primary modelling population.
- A chronological (time-based) train/test split, to evaluate the model on its ability to predict future appointments from past data.
- A logistic regression baseline, followed by tree-based ensemble candidates (random forest, gradient boosting).
- Evaluation using precision, recall, F1-score and ROC-AUC, rather than accuracy alone.

3. HealthConnect Resources Used

- HealthConnect Appointment Dataset (5,000 records) – the primary data source for all analysis.
- HealthConnect Data Dictionary – used to interpret each variable's definition and expected values.
- HealthConnect Clinic Knowledge Base – consulted specifically to confirm that a 'Cancelled' appointment reflects a proactive patient action, distinct from a silent no-show.

4. Key Assumptions and Limitations Identified

- The dataset is fictional and synthetic; findings are illustrative of methodology rather than clinically validated.
- The status of `waiting_time_minutes` was unresolved and flagged as a possible data-leakage risk pending clarification.
- The decision to exclude Cancelled appointments from the target definition was noted as sensitive to the clinic's actual operational priorities.
- Model fairness across demographic subgroups (gender, age) was identified as needing review before any real-world use.

5. What Was Proposed for Week 5

Resolve the `waiting_time_minutes` question with the data owner; complete a formal statistical and correlation review of the candidate features; finish categorical encoding and numerical scaling; and train and evaluate the proposed baseline model. This proposed scope is carried forward and assessed against what was actually delivered in the accompanying Week 5 Project Summary.